Om Prakash Choudhary

omprakashc@iisc.ac.in | Website | LinkedIn | GitHub

About Me

My research interests lie in **generative modeling**, with a focus on **reward alignment**. I am excited by the diverse applications of generative models and eager to contribute to their development.

Education

Indian Institute of Science, *B.Tech in Mathematics and Computing* Aug 2023 - Present

- GPA: 8.5 / 10.0
- **Medium of Instruction:** English
- **Relevant Courses:** Deep Generative Models, Machine Learning, Optimization Theory, Reinforcement Learning, Multi-armed bandit
- **Recognition:** Awarded ITTIAM BTech Fellowship

Publications

Om Prakash Choudhary, Dhruva Kashyap, Chiranjib Bhattacharyya.
Steer Early, Compose Better: Avoiding Late Horizon Steering by Optimal Residual Control for Diffusion Alignment.
Under review at ICLR 2027. [Manuscript]

Experience

Student Researcher, ML Lab, CSA, IISc May 2025 - Present
Advisor: Prof. Chiranjib Bhattacharyya

Steer Early, Compose Better [Manuscript] *September 2026*

- Developed a reward alignment method for diffusion models that enables alignment without a *value network*.
- Achieved up to **12**% improvement on compositional alignment tasks and up to **20**% lower detection rates for unlearned concepts compared with recent baselines.
- **Technology:** PyTorch, Accelerate, Wandb, Stable Diffusion, VLM.

Editing Concepts in Stable Diffusion [Blog] *July 2025*

- Developed a model editing approach for **Stable Diffusion** to unlearn specific concepts, as part of the **U&ME Challenge by ICCV 2025**.
- Achieved **15**% improvement in average forget score over diffusion-editing baselines in literature.
- **Technology:** Stable Diffusion, CLIP, PyTorch, Accelerate, Wandb.

Research Intern, HP Inc. May 2026 - August 2026

Multi Token Prediction

- Developed a training pipeline for a **multi-token draft model** to accelerate LLM inference, inspired by diffusion-based speculative decoding approaches **DFlash** and **DSpark**.
- **Technology:** PyTorch, Accelerate, Wandb, Qwen, vLLM.

Projects

Diffusion Quote Generator [HuggingFace Spaces] *January 2026*

- Pretrained a 149M-parameter **LLaDA** model on 10,000 books from Project Gutenberg.
- Supervised fine-tuning on a 500k-quote dataset.
- Applied 8-bit Quantization-Aware Training (QAT) for efficient deployment on HuggingFace Spaces.

Text-Conditioned Image Synthesis [GitHub] *May 2025*

- Implemented a **Latent Diffusion Model** with CLIP text encoder for conditional image generation.

Diffusion Models [GitHub] *March - April 2025*

- Studied influential papers on generative modeling, VAE, DDPM, DDIM, and Score-Based Generative Models.
- Implemented and benchmarked DDPM, DDIM, and Score-Based Models from scratch.

Neural Network From Scratch [GitHub] *June 2024*

- Built a minimal deep learning framework in NumPy with popular optimizers like SGD, Momentum, and Adam and benchmarked it on the MNIST dataset.